

Foundation (Jalapeno)

Q1. Solve the inequality $|x - 5| < 3$.

- (A) $2 < x < 8$
 (B) $2 \leq x \leq 8$
 (C) $0 < x < 10$
 (D) $-1 < x < 9$
 (E) $1 < x < 7$

Q2. Graph the solution set of $|x + 2| \geq 4$ on a number line.

- (A) $x \leq -6$ or $x \geq 2$
 (B) $x < -6$ or $x > 2$
 (C) $x \leq -2$ or $x \geq 6$
 (D) $x < -2$ or $x > 6$
 (E) $x \geq -6$ and $x \leq 2$

Q3. Express the solution set of $|3x - 2| > 5$ in interval notation.

- (A) $(-\infty, -1) \cup (1, \infty)$
 (B) $(-\infty, -1] \cup [1, \infty)$
 (C) $(-\infty, -$

$\frac{2}{3}) \cup (\frac{2}{3}, \infty)$

- (C) $(-\infty, \frac{2}{3}) \cup (\frac{2}{3}, \infty)$

- (C) $(-\infty, \frac{2}{3}] \cup [\frac{2}{3}, \infty)$

Q4. A stock's price changed by $|2x - 5|$ dollars. If the price change was between

2

and

8

, find the possible values of x .

(A)

$\frac{3}{2} < x < \frac{9}{2}$

- (A) $\frac{3}{2} \leq x \leq \frac{9}{2}$

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(A)

$\frac{1}{2} < x < \frac{7}{2}$

Q5. A savings account earns interest of $|0.05x - 100|$ dollars per year. Find the values of x for which the interest earned is between

20

and

50

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- (A) $1600 < x < 2800$
 (B) $1600 \leq x \leq 2800$
 (C) $1600 < x < 3000$
 (D) $1800 < x < 2600$
 (E) $2000 < x < 2400$

Q6. The tax on a stock investment is $|0.1x - 50|$ dollars. Find the values of x for which the tax paid is less than

20

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- (A) $100 < x < 700$
 (B) $100 \leq x \leq 700$
 (C) $300 < x < 500$
 (D) $200 < x < 600$
 (E) $0 < x < 400$

Q7. The distance between two cities is $|x - 200|$ miles. Find the values of x for which the distance is greater than 50 miles.

- (A) $x < 150$ or $x > 250$
 (B) $x \leq 150$ or $x \geq 250$
 (C) $x < 250$ or $x > 150$
 (D) $x \leq 250$ or $x \geq 150$
 (E) $150 < x < 250$

Q8. A company's profit is $|0.2x - 150|$ dollars. Find the values of x for which the profit is greater than

50

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- (A) $500 < x < 1000$
 (B) $500 \leq x \leq 1000$
 (C) $400 < x < 1200$
 (D) $1000 < x < 400$
 (E) $700 < x < 800$

Open Ended Questions (FRQ)

Q9. Model the situation where a stock's price increased by $|1.5x - 20|$ dollars and decreased by $|0.8x - 15|$ dollars. Find the values of x for which the net change in the stock's price is between

10

and

30

. Provide your solution in interval notation.

Q10. A bank account earns $|0.03x - 10|$ dollars interest per month, but is charged $|0.02x - 5|$ dollars in fees. Find the values of x for which the net interest earned is greater than

. Graph your solution set on a number line and express it in interval notation.

Chef's Tips

- Ensure students understand the concept of absolute value inequalities.
- Remind students to consider both cases when solving absolute value inequalities.
- Emphasize the importance of expressing solutions in interval notation and graphing on a number line.